

Element C:

Design Requirement (In order from greatest to least)	Source	In-Depth Analysis	Units Used
Educational Value	link	Aquaponics offers a platform for teaching various science and math related classes, such as biology, chemistry, and sustainable agriculture. It also provides hands-on learning experiences for students.	Measurable by finding the average scores in science classes and then comparing them to average scores after the aquaponics system was built.
Scale and Size	link	Determining an appropriate scale and size ensures the system is manageable for maintenance and operation within the school setting. It affects factors such as water volume and nutrient cycling efficiency. A good thing to know is the more water volume there is, the easier it is to maintain.	Dimensions (e.g., m ²)

Safety Considerations	link	Safety considerations include preventing accidents such as shocks from loose wires, slipping on water that spilled out from the system, as well as minimizing risks associated with water quality, equipment failure, and the chemicals used to clean the system.	Measuring the amount of injuries that occurred due to the aquaponics project.
Accessibility	link	Ensuring accessibility means designing the system to work for students of all abilities, including those with physical disabilities. This involves considerations such as the height of the system and ease of use.	Dimensions (e.g., cm)
Sustainability	link	Sustainability is about practices that conserve resources, minimize environmental impact, and promote long term viability. This includes efficient water use and energy conservation.	Measurable by the amount of times that resources needed to be replaced.
Monitoring and Control	link	Control systems enable students to track key factors such as water quality, temperature, pH, and nutrient levels. This allows for a deeper understanding of the aquaponics system and allows for adjustments to increase the performance of the system.	Various (e.g., pH, °C)
Integration with Curriculum	link	Integrating aquaponics into the school's curriculum provides learning opportunities across science, technology, engineering, and math. It enhances student engagement and is relevant to real world issues relating to farming waste.	Measurable by finding the average scores in science

			classes and then comparing them to average scores after the aquaponics system was built.
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C6. Primary stakeholder validation. There is evidence that requirements represent the needs of, and have been validated by, _Clint Rickert, T&E - CMS_____ primary stakeholder groups.

Mr. Rickert - The group has completed all possible requirements for this school year in the construction and preliminary setup of the system. Post project paperwork has been submitted to Millersville University and is awaiting final approval. There will be more opportunities for those who are interested to continue to work on the system next school year. Particularly in making it fit into the district provided enclosure and possibly the development of an automated monitoring/treatment system that makes it possible to manage the system via a cell phone app.